

name, t, x, v
stamp, i
m

		val $v_i : T$	Cbinders = v_i	value specification
		C/p	M	strengthening
		(C)	M	
T	::=			Types
		t_i		lexical type reference
		$p.t$		structure-based type reference
		$T_1 \rightarrow T_2$		function type
		$T\{n_i \leftarrow p.n n_i \in BV(D)\}$	M	substitution
		(T)	M	
p	::=			Structure paths
		x_i		lexical structure reference
		$p.x$		structure-based structure reference
E	::=			Typing contexts
		$E; C$		
		$E_1; C; E_2$	M	
		$E; \mathbf{rev} D$	M	
M	::=			Terms
		v_i		Lexical term reference
		$p.v$		Structure-based term reference
		$\lambda v_i : T. M$	bind v_i in M	Function expression
		$M_1 M_2$		Function application
		let d in M		Lexical binding
		$M\{v_i \leftarrow V\}$	M	substitution
		$M\{n_i \leftarrow p.n n_i \in BV(d)\}$	M	substitution
		(M)	M	
V	::=			Values
		$\lambda v_i : T. M$		
cV	::=			Definition values
		type $t_i = T$		
		datatype t_i		
		structure $x_i = sV$		
		val $v_i = V$		
sV	::=			Module values
		struct dV end		
		functor $(x_i : S)s$		
dV	::=			Structure body values
		ϵ		
		$cV; dV$		
Δ	::=			Value environments

		$\Delta; c$ $\Delta_1; c; \Delta_2$ $\Delta; \mathbf{rev} d$	 M M
<i>terminals</i>	$::=$	$\rightarrow$ $\vdash$ $\approx$ $\leftarrow$ $\mapsto$ λ $\rightsquigarrow$	
<i>formula</i>	$::=$	$judgement$ $x_i \notin BV(E)$	
<i>pathE</i>	$::=$	$E \vdash p \mapsto S$	
<i>WFtype</i>	$::=$	$E \vdash T \mathbf{type}$	
<i>WFSig</i>	$::=$	$E \vdash D \mathbf{decl}$ $E \vdash S \mathbf{signature}$	
<i>eq</i>	$::=$	$E \vdash T \approx T'$	
<i>sub</i>	$::=$	$E \vdash S <: S'$ $E \vdash D <: D'$ $E \vdash C <: C'$	
<i>typ</i>	$::=$	$E \vdash s : S$ $E \vdash c : C$ $E \vdash d : D$ $E \vdash M : T$	
<i>pathDelta</i>	$::=$	$\Delta \vdash p \mapsto s$	
<i>reduce</i>	$::=$	$\Delta \vdash M \rightsquigarrow M'$ $\Delta \vdash c \rightsquigarrow c'$ $\Delta \vdash d \rightsquigarrow d'$	

$$\begin{array}{l|l} & \Delta \vdash s \rightsquigarrow s' \\ \hline \text{deltaE} & ::= \\ & | \vdash \Delta : E \end{array}$$

$$\begin{array}{l|l} \text{judgement} & ::= \\ & | \text{pathE} \\ & | \text{WFtype} \\ & | \text{WFsig} \\ & | \text{eq} \\ & | \text{sub} \\ & | \text{typ} \\ & | \text{pathDelta} \\ & | \text{reduce} \\ & | \text{deltaE} \end{array}$$

$$\begin{array}{l|l} \text{user_syntax} & ::= \\ & | \text{name} \\ & | \text{stamp} \\ & | m \\ & | \text{ident} \\ & | s \\ & | d \\ & | c \\ & | S \\ & | D \\ & | C \\ & | T \\ & | p \\ & | E \\ & | M \\ & | V \\ & | cV \\ & | sV \\ & | dV \\ & | \Delta \\ & | \text{terminals} \\ & | \text{formula} \end{array}$$

$E \vdash p \mapsto S$

$$\frac{}{E_1; \mathbf{structure} \, x_i : S; E_2 \vdash x_i \mapsto S/x_i} \text{PATH E1}$$

$$\frac{E \vdash p \mapsto \mathbf{sig} \, D_1; \mathbf{structure} \, x_i : S; D_2 \mathbf{end}}{E \vdash p.x \mapsto S\{n_i \leftarrow p.n | n_i \in BV(D_1)\}} \text{PATH E2}$$

$E \vdash T \mathbf{type}$

$$\frac{}{E_1; \mathbf{type} \, t_i; E_2 \vdash t_i \mathbf{type}} \text{WFTYPE1}$$

$$\frac{}{E_1; \mathbf{type} \, t_i = T; E_2 \vdash t_i \mathbf{type}} \text{WFTYPE2}$$

$$\frac{E \vdash p \mapsto \mathbf{sig} D_1; \mathbf{type} t_i; D_2 \mathbf{end}}{E \vdash p.t \mathbf{type}} \quad \text{WFTYPE3}$$

$$\frac{E \vdash p \mapsto \mathbf{sig} D_1; \mathbf{type} t_i = T; D_2 \mathbf{end}}{E \vdash p.t \mathbf{type}} \quad \text{WFTYPE4}$$

$$\frac{\begin{array}{c} E \vdash T_1 \mathbf{type} \\ E \vdash T_2 \mathbf{type} \end{array}}{E \vdash (T_1 \rightarrow T_2) \mathbf{type}} \quad \text{WFTYPE5}$$

$$\boxed{E \vdash D \mathbf{decl}}$$

$$\overline{E \vdash \epsilon \mathbf{decl}} \quad \text{WFDECL1}$$

$$\frac{\begin{array}{c} E \vdash T \mathbf{type} \\ E; \mathbf{type} t_i = T \vdash D \mathbf{decl} \end{array}}{E \vdash (\mathbf{type} t_i = T; D) \mathbf{decl}} \quad \text{WFDECL2}$$

$$\frac{E; \mathbf{type} t_i \vdash D \mathbf{decl}}{E \vdash (\mathbf{type} t_i; D) \mathbf{decl}} \quad \text{WFDECL3}$$

$$\frac{\begin{array}{c} E \vdash S \mathbf{signature} \\ E; \mathbf{structure} x_i : S \vdash D \mathbf{decl} \end{array}}{E \vdash (\mathbf{structure} x_i : S; D) \mathbf{decl}} \quad \text{WFDECL4}$$

$$\frac{\begin{array}{c} E \vdash T \mathbf{type} \\ E; \mathbf{val} v_i : T \vdash D \mathbf{decl} \end{array}}{E \vdash (\mathbf{val} v_i : T; D) \mathbf{decl}} \quad \text{WFDECL5}$$

$$\boxed{E \vdash S \mathbf{signature}}$$

$$\frac{E \vdash D \mathbf{decl}}{E \vdash (\mathbf{sig} D \mathbf{end}) \mathbf{signature}} \quad \text{WFSIG1}$$

$$\frac{\begin{array}{c} E \vdash S_1 \mathbf{signature} \\ E; \mathbf{structure} x_i : S_1 \vdash S_2 \mathbf{signature} \end{array}}{E \vdash (\mathbf{functor} (x_i : S_1) S_2) \mathbf{signature}} \quad \text{WFSIG2}$$

$$\boxed{E \vdash T \approx T'}$$

$$\overline{E_1; \mathbf{type} t_i = T; E_2 \vdash t_i \approx T} \quad \text{EQT1}$$

$$\frac{E \vdash p \mapsto \mathbf{sig} D_1; \mathbf{type} t_i = T; D_2 \mathbf{end}}{E \vdash p.t \approx T\{n_i \leftarrow p.n | n_i \in BV(D_1)\}} \quad \text{EQT2}$$

$$\overline{E \vdash T \approx T} \quad \text{EQT3}$$

$$\frac{E \vdash T_2 \approx T_1}{E \vdash T_1 \approx T_2} \quad \text{EQT4}$$

$$\frac{\begin{array}{c} E \vdash T_1 \approx T_2 \\ E \vdash T_2 \approx T_3 \end{array}}{E \vdash T_1 \approx T_3} \quad \text{EQT5}$$

$$\frac{\begin{array}{c} E \vdash T_1 \approx T_3 \\ E \vdash T_2 \approx T_4 \end{array}}{E \vdash (T_1 \rightarrow T_2) \approx (T_3 \rightarrow T_4)} \quad \text{EQT6}$$

$$\boxed{E \vdash S <: S'}$$

$$\frac{E \vdash S_2 <: S_1 \quad E; \mathbf{structure} \, x_i : S_2 \vdash S'_1 <: S'_2}{E \vdash \mathbf{functor} \, (x_i : S_1) S'_1 <: \mathbf{functor} \, (x_i : S_2) S'_2} \text{ SUBS1}$$

$$\frac{E \vdash D <: D'}{E \vdash \mathbf{sig} \, D \mathbf{end} <: \mathbf{sig} \, D' \mathbf{end}} \text{ SUBS2}$$

$$\boxed{E \vdash D <: D'}$$

$$\overline{E \vdash \epsilon <: \epsilon} \text{ SUBD1}$$

$$\frac{E \vdash C <: C' \quad E; C \vdash D <: D'}{E \vdash C; D <: C'; D'} \text{ SUBD2}$$

$$\frac{E; C \vdash D <: D'}{E \vdash C; D <: D'} \text{ SUBD3}$$

$$\boxed{E \vdash C <: C'}$$

$$\overline{E \vdash (\mathbf{type} \, t_i) <: (\mathbf{type} \, t_i)} \text{ SUBC1}$$

$$\overline{E \vdash (\mathbf{type} \, t_i = T) <: (\mathbf{type} \, t_i)} \text{ SUBC2}$$

$$\frac{E \vdash T_1 \approx T_2}{E \vdash (\mathbf{type} \, t_i = T_1) <: (\mathbf{type} \, t_i = T_2)} \text{ SUBC4}$$

$$\frac{E \vdash S_1 <: S_2}{E \vdash (\mathbf{structure} \, x_i : S_1) <: (\mathbf{structure} \, x_i : S_2)} \text{ SUBC5}$$

$$\frac{E \vdash T_1 \approx T_2}{E \vdash (\mathbf{val} \, v_i : T_1) <: (\mathbf{val} \, v_i : T_2)} \text{ SUBC6}$$

$$\boxed{E \vdash s : S}$$

$$\frac{E \vdash p \mapsto S}{E \vdash p : S} \text{ TYPE_S1}$$

$$\frac{E \vdash S \mathbf{signature} \quad x_i \notin BV(E) \quad E; \mathbf{structure} \, x_i : S \vdash s : S'}{E \vdash \mathbf{functor} \, (x_i : S) s : \mathbf{functor} \, (x_i : S) S'} \text{ TYPE_S2}$$

$$\frac{E \vdash s : \mathbf{functor} \, (x_i : S') S \quad E \vdash p \mapsto S'' \quad E \vdash S'' <: S'}{E \vdash s(p) : S\{x_i \leftarrow p\}} \text{ TYPE_S3}$$

$$\frac{E \vdash d : D}{E \vdash \mathbf{struct} \, d \mathbf{end} : \mathbf{sig} \, D \mathbf{end}} \text{ TYPE_S4}$$

$$\frac{E \vdash S_1 \mathbf{signature} \quad E \vdash s : S_2 \quad E \vdash S_2 <: S_1}{E \vdash s :> S_1 : S_1} \text{ TYPE_S5}$$

$$\boxed{E \vdash c : C}$$

$$\frac{E \vdash T \text{ type} \quad t_i \notin BV(E)}{E \vdash \text{type } t_i = T : \text{type } t_i = T} \quad \text{TYPE_C1}$$

$$\frac{t_i \notin BV(E)}{E \vdash \text{datatype } t_i : \text{type } t_i} \quad \text{TYPE_C2}$$

$$\frac{E \vdash s : S \quad x_i \notin BV(E)}{E \vdash \text{structure } x_i = s : \text{structure } x_i : S} \quad \text{TYPE_C3}$$

$$\frac{E \vdash M : T \quad v_i \notin BV(E)}{E \vdash \text{val } v_i = M : \text{val } v_i : T} \quad \text{TYPE_C4}$$

$$\boxed{E \vdash d : D}$$

$$\overline{E \vdash \epsilon : \epsilon} \quad \text{TYPE_D1}$$

$$\frac{E \vdash c : C \quad E; C \vdash d : D}{E \vdash c; d : C; D} \quad \text{TYPE_D2}$$

$$\boxed{E \vdash M : T}$$

$$\overline{E_1; \text{val } v_i : T; E_2 \vdash v_i : T} \quad \text{TYPE_M1}$$

$$\frac{E \vdash p \mapsto \text{sig } D_1; \text{val } v_i : T; D_2 \text{ end}}{E \vdash p.v : T\{n_i \leftarrow p.n \mid n_i \in BV(D_1)\}} \quad \text{TYPE_M2}$$

$$\frac{v_i \notin BV(E) \quad E \vdash T_1 \text{ type} \quad E; \text{val } v_i : T_1 \vdash M : T_2}{E \vdash \lambda v_i : T_1. M : T_1 \rightarrow T_2} \quad \text{TYPE_M3}$$

$$\frac{E \vdash M_1 : T_1 \rightarrow T_2 \quad E \vdash M_2 : T_1}{E \vdash M_1 M_2 : T_2} \quad \text{TYPE_M4}$$

$$\frac{E \vdash d : D \quad E; \text{rev } D \vdash M : T_1 \quad E; \text{rev } D \vdash T_1 \approx T_2 \quad E \vdash T_2 \text{ type}}{E \vdash \text{let } d \text{ in } M : T_2} \quad \text{TYPE_M5}$$

$$\boxed{\Delta \vdash p \mapsto s}$$

$$\overline{\Delta_1; \text{structure } x_i = s; \Delta_2 \vdash x_i \mapsto s/x_i} \quad \text{PATH_DELTA1}$$

$$\frac{\Delta \vdash p \mapsto \text{struct } d_1; \text{structure } x_i = s; d_2 \text{ end}}{\Delta \vdash p.x \mapsto s\{d_1 \leftarrow p\}} \quad \text{PATH_DELTA2}$$

$$\boxed{\Delta \vdash M \rightsquigarrow M'}$$

$$\begin{array}{c}
\frac{}{\Delta_1; \mathbf{val} v_i = V; \Delta_2 \vdash v_i \rightsquigarrow V} \text{REDMVAR} \\
\frac{\Delta \vdash p \mapsto \mathbf{struct} dV_1; \mathbf{val} v_i = V; dV_2 \mathbf{end}}{\Delta \vdash p.v \rightsquigarrow V\{n_i \leftarrow p.n \mid n_i \in BV(dV_1)\}} \text{REDMPATH} \\
\frac{}{\Delta \vdash (\lambda v_i : T.M) V \rightsquigarrow M\{v_i \leftarrow V\}} \text{REDMAPP} \\
\frac{}{\Delta \vdash \mathbf{let} dV \mathbf{in} (\lambda v_i : T.M) \rightsquigarrow \lambda v_i : T. \mathbf{let} dV \mathbf{in} M} \text{REDMLET} \\
\frac{\Delta \vdash M_1 \rightsquigarrow M_3}{\Delta \vdash M_1 M_2 \rightsquigarrow M_3 M_2} \text{REDMAPP_CTXT1} \\
\frac{\Delta \vdash M_1 \rightsquigarrow M_2}{\Delta \vdash V M_1 \rightsquigarrow V M_2} \text{REDMAPP_CTXT2} \\
\frac{\Delta \vdash d_1 \rightsquigarrow d_2}{\Delta \vdash \mathbf{let} d_1 \mathbf{in} M \rightsquigarrow \mathbf{let} d_2 \mathbf{in} M} \text{REDMLET_CTXT1} \\
\frac{\Delta; \mathbf{rev} dV \vdash M_1 \rightsquigarrow M_2}{\Delta \vdash \mathbf{let} dV \mathbf{in} M_1 \rightsquigarrow \mathbf{let} dV \mathbf{in} M_2} \text{REDMLET_CTXT2}
\end{array}$$

$$\boxed{\Delta \vdash c \rightsquigarrow c'}$$

$$\begin{array}{c}
\frac{\Delta \vdash s_1 \rightsquigarrow s_2}{\Delta \vdash \mathbf{structure} x_i = s_1 \rightsquigarrow \mathbf{structure} x_i = s_2} \text{REC_C_STR_CTXT} \\
\frac{\Delta \vdash M_1 \rightsquigarrow M_2}{\Delta \vdash \mathbf{val} v_i = M_1 \rightsquigarrow \mathbf{val} v_i = M_2} \text{REC_C_VAL_CTXT}
\end{array}$$

$$\boxed{\Delta \vdash d \rightsquigarrow d'}$$

$$\begin{array}{c}
\frac{\Delta \vdash c_1 \rightsquigarrow c_2}{\Delta \vdash c_1; d \rightsquigarrow c_2; d} \text{RED_D_D_CTXT1} \\
\frac{\Delta; cV \vdash d_1 \rightsquigarrow d_2}{\Delta \vdash cV; d_1 \rightsquigarrow cV; d_2} \text{RED_D_D_CTXT2}
\end{array}$$

$$\boxed{\Delta \vdash s \rightsquigarrow s'}$$

$$\begin{array}{c}
\frac{\Delta \vdash p \mapsto sV}{\Delta \vdash p \rightsquigarrow sV} \text{RED_S_PATH} \\
\frac{}{\Delta \vdash (\mathbf{functor} (x_i : S)s)(p) \rightsquigarrow s\{x_i \leftarrow p\}} \text{RED_S_FTORAPP} \\
\frac{}{\Delta \vdash s :> S \rightsquigarrow s} \text{RED_S_SEAL} \\
\frac{\Delta \vdash d_1 \rightsquigarrow d_2}{\Delta \vdash \mathbf{struct} d_1 \mathbf{end} \rightsquigarrow \mathbf{struct} d_2 \mathbf{end}} \text{RED_S_STR_CTXT} \\
\frac{\Delta \vdash s_1 \rightsquigarrow s_2}{\Delta \vdash s_1(p) \rightsquigarrow s_2(p)} \text{RED_S_FTORAPP_CTXT}
\end{array}$$

$$\boxed{\vdash \Delta : E}$$

$$\begin{array}{c}
\frac{}{\vdash:} \text{DELTA E1} \\
\frac{\vdash \Delta : E \quad E \vdash cV : C}{\vdash \Delta; cV : E; C} \text{DELTA E2}
\end{array}$$

Definition rules: 67 good 0 bad
Definition rule clauses: 142 good 0 bad